

Rapport Technique : Installation et Configuration d'Hadoop sur Windows

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Installation de Hadoop sur Windows — Prérequis

1

Java JDK

Hadoop nécessite un environnement JVM opérationnel ; nous privilégions l'utilisation d'OpenJDK 8 ou Java 8 pour assurer une compatibilité optimale.

2

Windows Terminal

Nous utilisons PowerShell ou Command Prompt pour l'exécution des processus et l'édition des fichiers de configuration.

3

Windows 10/11

Un système d'exploitation Windows moderne est requis pour assurer la stabilité et la compatibilité avec les outils Hadoop.

4

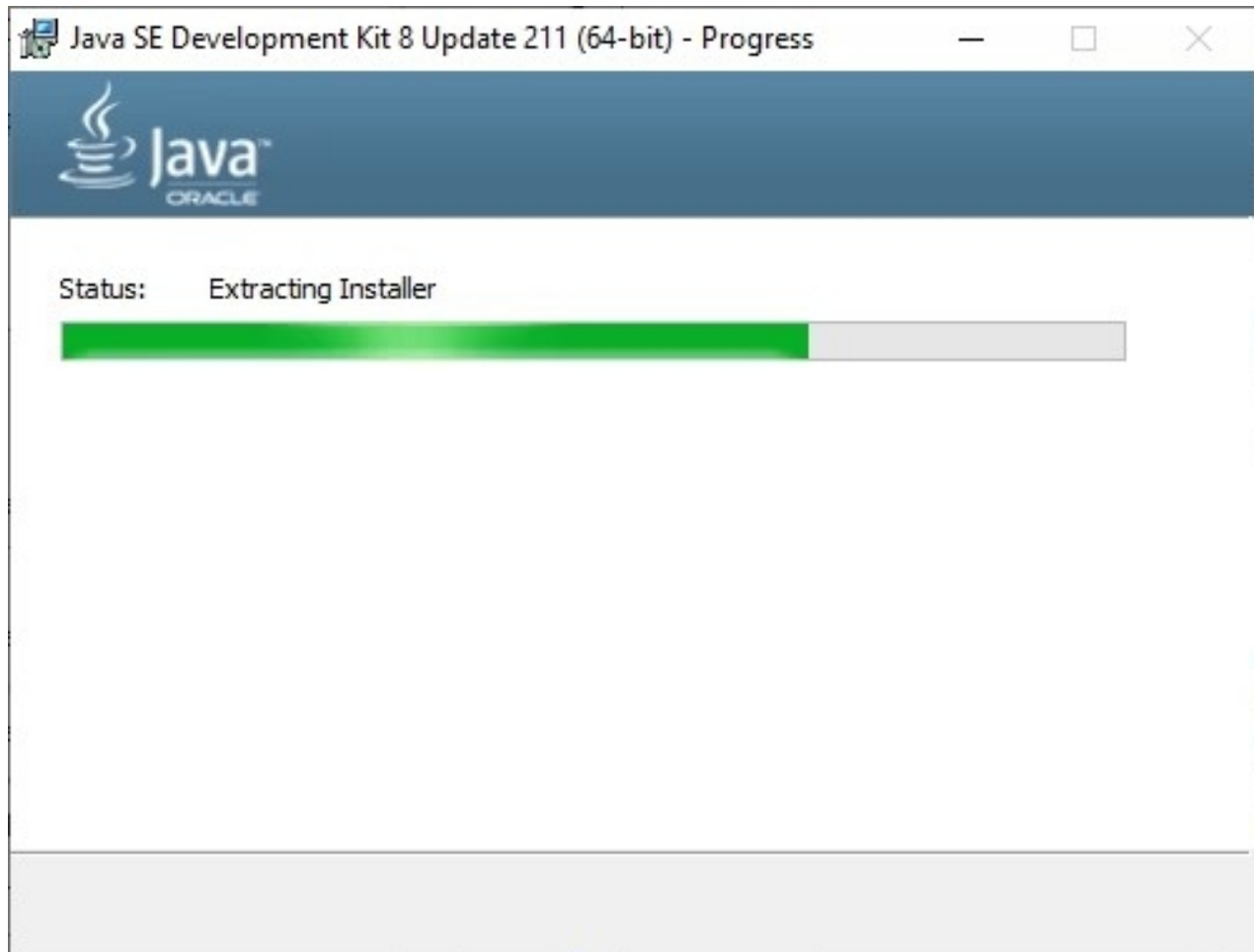
Connexion Internet

Une connectivité réseau est requise pour procéder au téléchargement sécurisé des archives Hadoop ainsi que des dépendances système nécessaires.

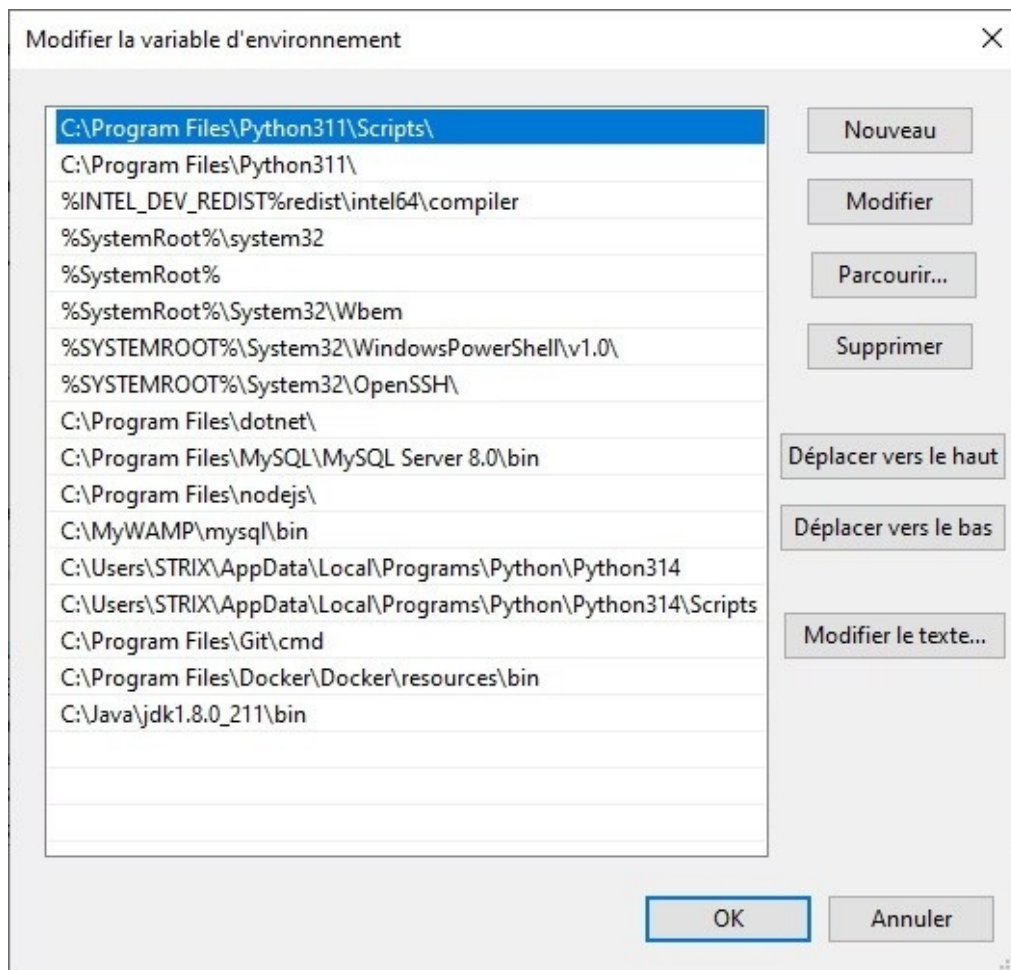
Installation de Java

Pour installer Java sur Windows, suivez ces étapes afin d'installer OpenJDK 8 et de configurer votre environnement :

1. Téléchargement et l'installation

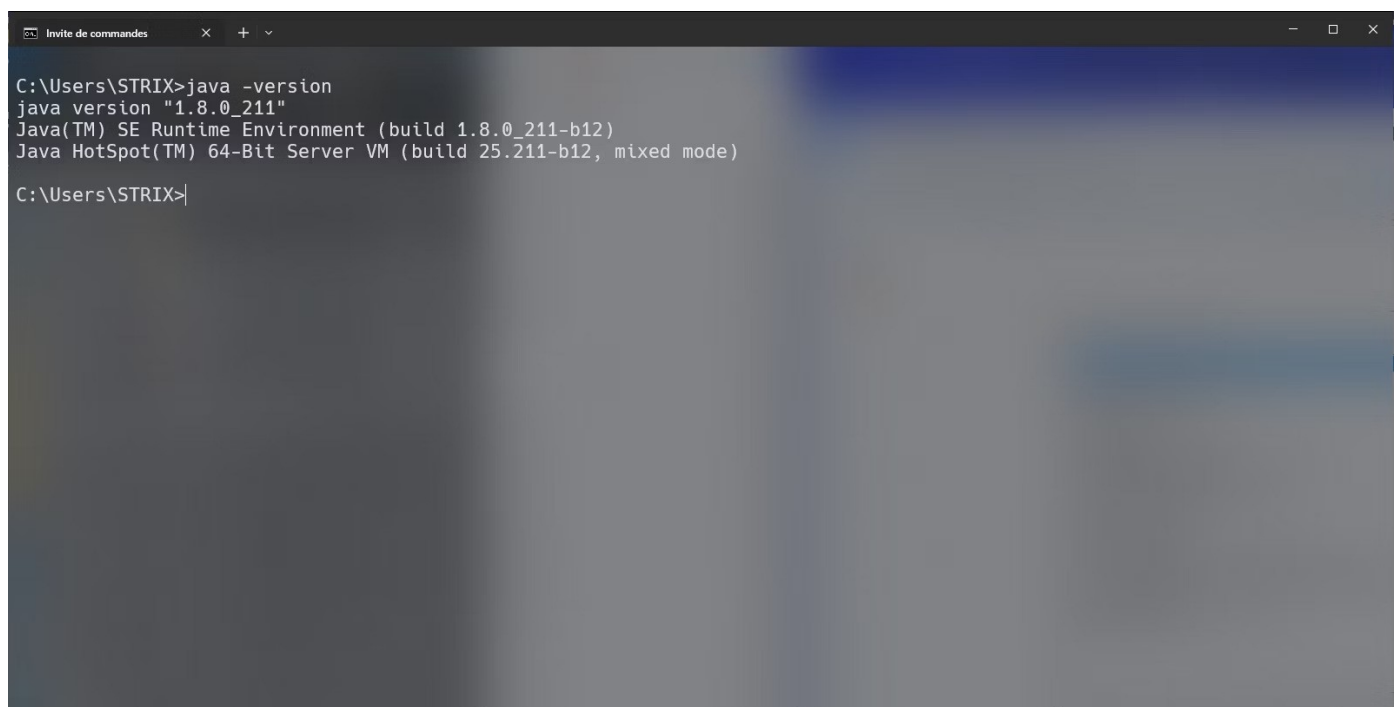


2. Définir les Variables d'Environnement



3. Vérification d'installation

```
java -version
```

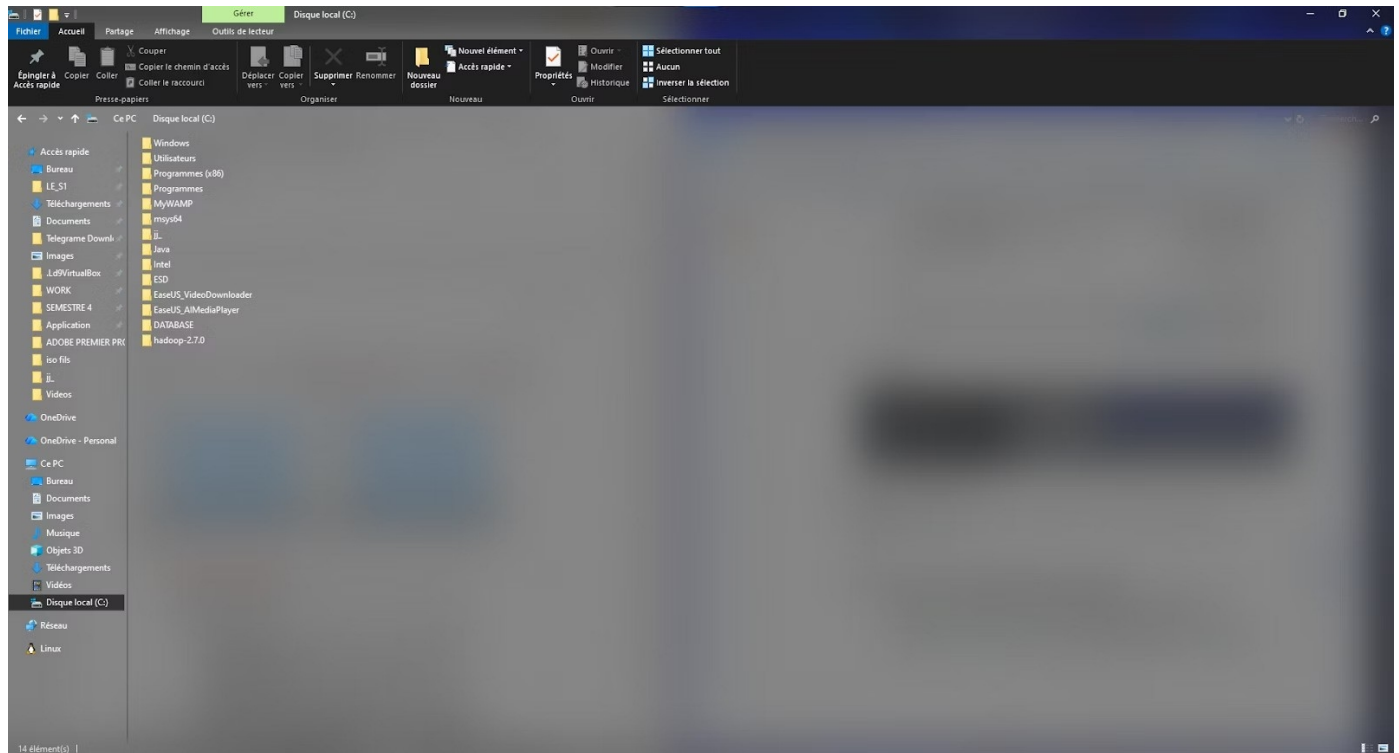


Préparation de Système Hadoop Windows

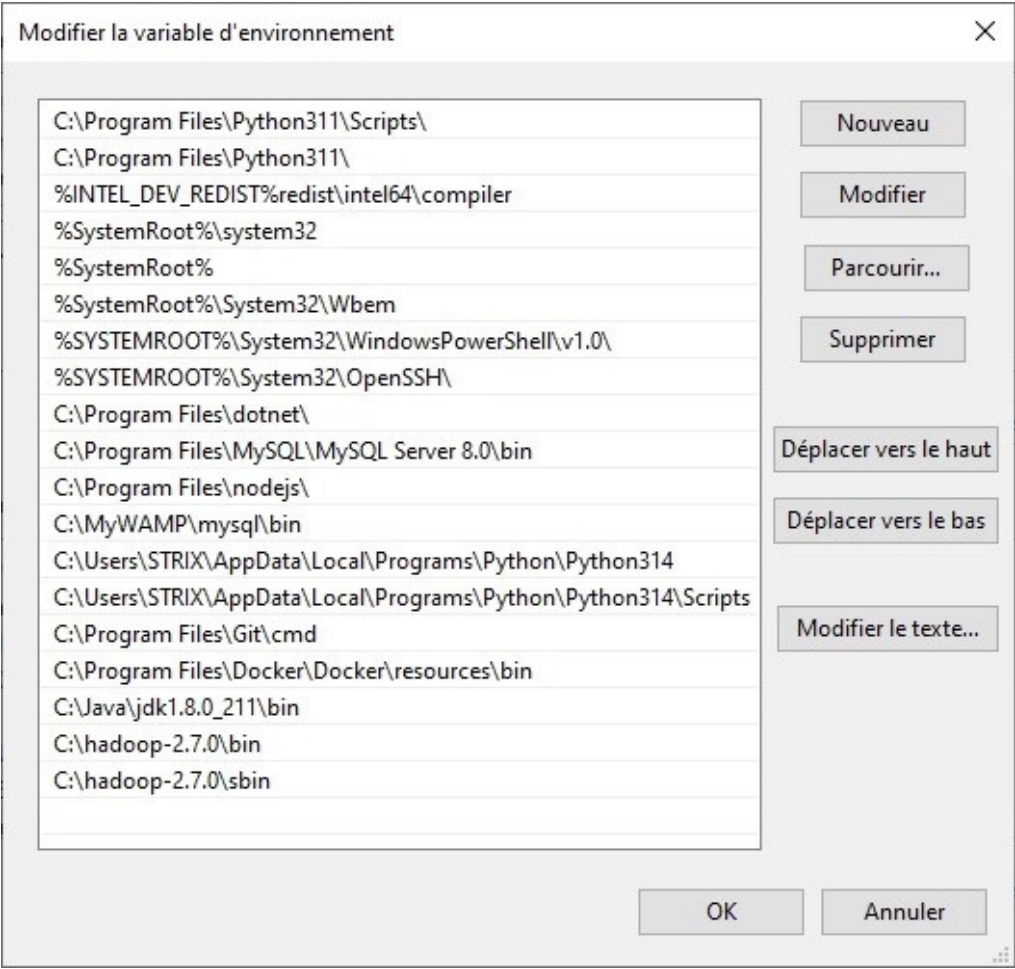
1. Téléchargement et Extraction de fichier Hadoop

Extraction du contenu de l'archive dans un répertoire facile d'accès, comme

C:\hadoop



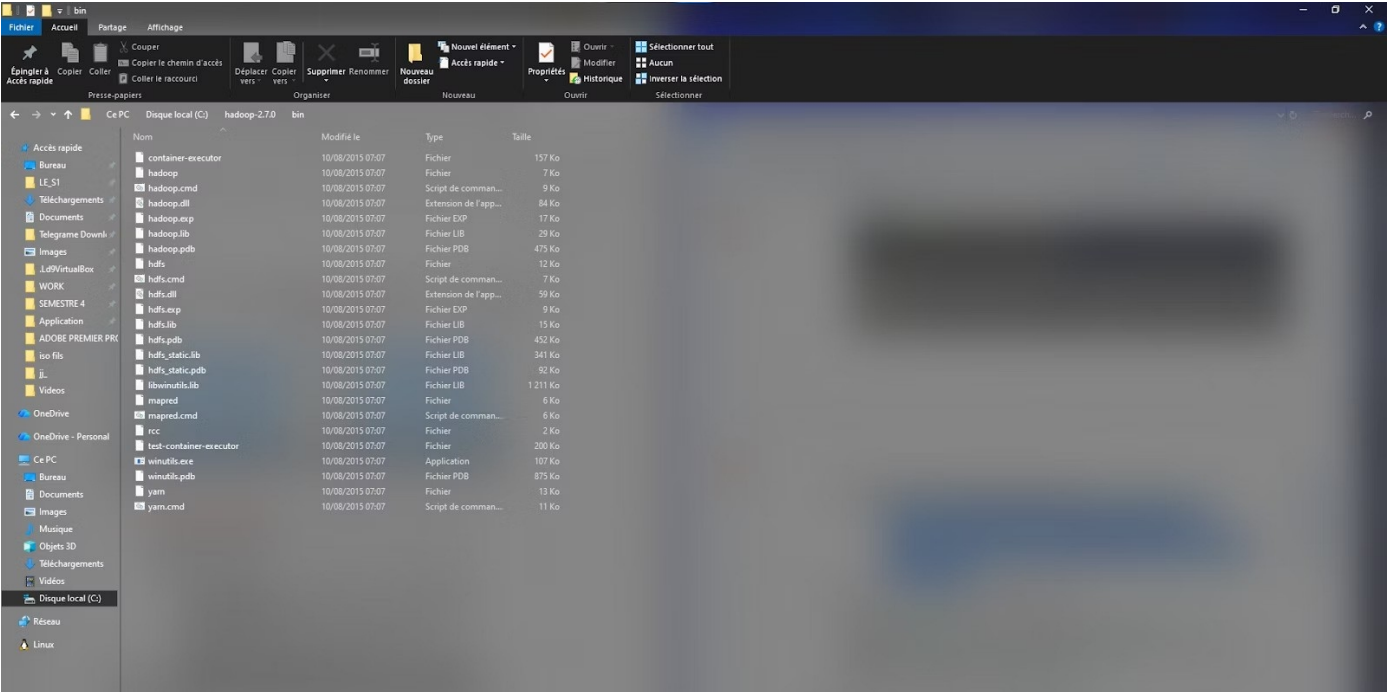
2. Définir les Variables d'Environnement



3. Intégration des Fichiers Binaires

Déplacez le fichier bin du hadooponwindows-master téléchargé dans le répertoire bin de votre installation Hadoop

C:\hadoop\bin

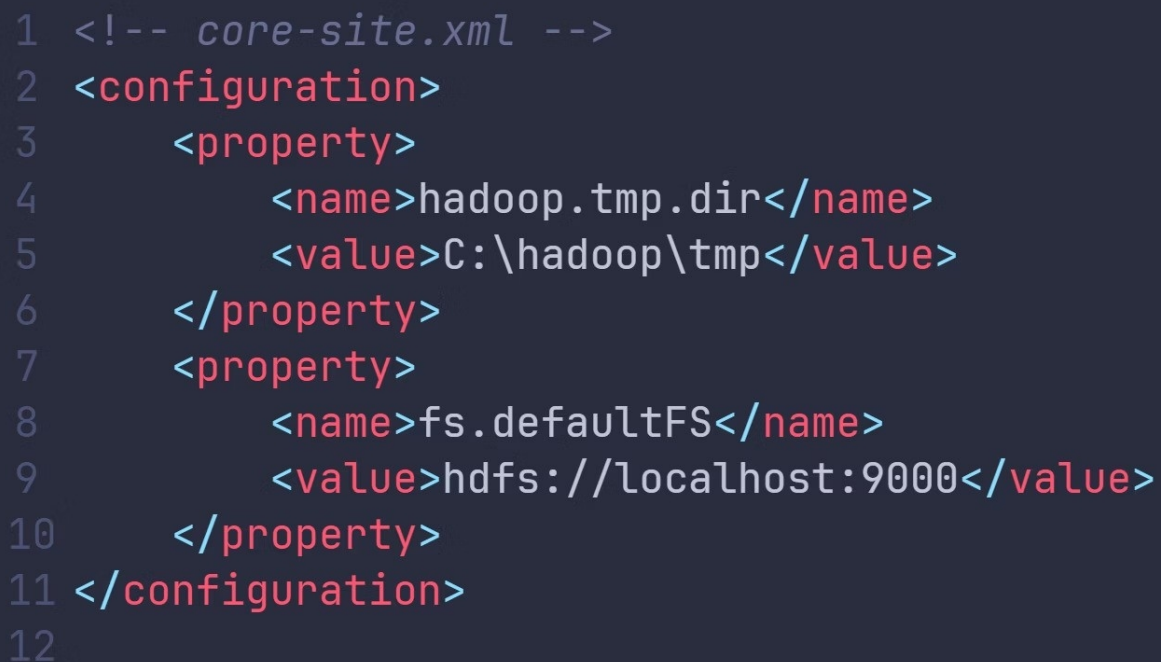


Configuration des Fichiers Hadoop

Pour assurer le bon fonctionnement d'Hadoop sur Windows, il est nécessaire de configurer manuellement les variables système. Suivez ces étapes pour configurer JAVA_HOME, HADOOP_HOME et mettre à jour le PATH.

1. Configuration de core-site.xml

Ce fichier configure le système de fichiers par défaut (HDFS) pour Hadoop



```
1 <!-- core-site.xml -->
2 <configuration>
3   <property>
4     <name>hadoop.tmp.dir</name>
5     <value>C:\hadoop\tmp</value>
6   </property>
7   <property>
8     <name>fs.defaultFS</name>
9     <value>hdfs://localhost:9000</value>
10  </property>
11 </configuration>
12
```

2. Configuration de hdfs-site.xml

Ce fichier spécifie les emplacements des répertoires de données du NameNode et du DataNode.

```
1 <!-- hdfs-site.xml -->
2 <configuration>
3     <property>
4         <name>dfs.replication</name>
5         <value>1</value>
6     </property>
7     <property>
8         <name>dfs.namenode.name.dir</name>
9         <value>/C:/hadoop/data/nameNode</value>
10    </property>
11    <property>
12        <name>dfs.datanode.data.dir</name>
13        <value>/C:/hadoop/data/dataNode</value>
14    </property>
15 </configuration>
16
```

3. Configuration de mapred-site.xml

Ce fichier configure le framework d'exécution MapReduce pour utiliser YARN.

```
1 <!-- mapred-site.xml -->
2 <configuration>
3     <property>
4         <name>mapreduce.framework.name</name>
5         <value>yarn</value>
6     </property>
7 </configuration>
```


4. Configuration de yarn-site.xml

Ce fichier configure YARN (Yet Another Resource Negotiator), le gestionnaire de ressources de Hadoop.

```
1 <!-- yarn-site.xml -->
2 <configuration>
3   <property>
4     <name>yarn.nodemanager.aux-services</name>
5     <value>mapreduce_shuffle</value>
6   </property>
7
8   <property>
9     <name>yarn.nodemanager.aux-services.mapreduce.shuffle.class</name>
10    <value>org.apache.hadoop.mapred.ShuffleHandler</value>
11  </property>
12 </configuration>
```

5. Configuration de hadoop-env.cmd

Définir le chemin vers votre installation Java (JDK) en mettant à jour la variable :

```
1 # The java implementation to use.
2 export JAVA_HOME="C:\Java\jdk1.8.0_211"
```

Initialisation et Démarrage de Hadoop

1. Formatage du NameNode

Formatage du système de fichiers HDFS :

```
Administrateur : Invite de commandes
26/03/23 21:58:57 INFO namenode.FSDirectory: XAttrs enabled? true
26/03/23 21:58:57 INFO namenode.FSDirectory: Maximum size of an xattr: 16384
26/03/23 21:58:57 INFO namenode.NameNode: Caching file names occurring more than 10 times
26/03/23 21:58:57 INFO util.GSet: Computing capacity for map cachedBlocks
26/03/23 21:58:57 INFO util.GSet: VM type = 64-bit
26/03/23 21:58:57 INFO util.GSet: 0.25% max memory 889 MB = 2.2 MB
26/03/23 21:58:57 INFO util.GSet: capacity = 2^18 = 262144 entries
26/03/23 21:58:57 INFO namenode.FSNamesystem: dfs.namenode.safemode.threshold-pct = 0.9990000128746033
26/03/23 21:58:57 INFO namenode.FSNamesystem: dfs.namenode.safemode.min.datanodes = 0
26/03/23 21:58:57 INFO namenode.FSNamesystem: dfs.namenode.safemode.extension = 30000
26/03/23 21:58:57 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.window.num.buckets = 10
26/03/23 21:58:57 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.num.users = 10
26/03/23 21:58:57 INFO metrics.TopMetrics: NNTop conf: dfs.namenode.top.windows.minutes = 1,5,25
26/03/23 21:58:57 INFO namenode.FSNamesystem: Retry cache on namenode is enabled
26/03/23 21:58:57 INFO namenode.FSNamesystem: Retry cache will use 0.03 of total heap and retry cache entry expiry time is 600000 millis
26/03/23 21:58:57 INFO util.GSet: Computing capacity for map NameNodeRetryCache
26/03/23 21:58:57 INFO util.GSet: VM type = 64-bit
26/03/23 21:58:57 INFO util.GSet: 0.029999999329447746% max memory 889 MB = 273.1 KB
26/03/23 21:58:57 INFO util.GSet: capacity = 2^15 = 32768 entries
26/03/23 21:58:57 INFO namenode.FSImage: Allocated new BlockPoolId: BP-1336481334-192.168.1.7-1774299537323
26/03/23 21:58:57 INFO common.Storage: Storage directory C:\hadoop\data\namenode has been successfully formatted.
26/03/23 21:58:57 INFO namenode.NNStorageRetentionManager: Going to retain 1 images with txid >= 0
26/03/23 21:58:57 INFO util.ExitUtil: Exiting with status 0
26/03/23 21:58:57 INFO namenode.NameNode: SHUTDOWN_MSG:
/*****
SHUTDOWN_MSG: Shutting down NameNode at DESKTOP-BACH1KC/192.168.1.7
*****/
C:\Windows\system32>
```

2. Demarage des Services Hadoop

Start-dfs.cmd

```
Apache Hadoop Distribution - hadoop datanode
DEPRECATED: Use of this script to execute hdfs command is deprecated.
Instead use the hdfs command for it.
26/03/23 22:03:48 INFO datanode.DataNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting DataNode
STARTUP_MSG: host = DESKTOP-BACH1KC/192.168.1.7
STARTUP_MSG: args = []
STARTUP_MSG: version = 2.7.0
STARTUP_MSG: classpath = C:\hadoop-2.7.0\etc\hadoop;C:\hadoop-2.7.0\share\hadoop\common\lib\activation-1.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\apacheds-i18n-2.0.0-M15.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\apacheds-kerberos-codec-2.0.0-M15.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\api-asn1-api-1.0.0-M20.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\api-util-1.0.0-M20.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\asm-3.2.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\avro-1.7.4.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-beanutils-1.7.0.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-beanutils-core-1.8.0.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-cli-1.2.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-codec-1.4.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-collections-3.2.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-compress-1.4.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-configuration-1.6.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-digester-1.8.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-httpclient-3.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-io-2.4.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-lang-2.6.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-logging-1.1.3.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-math3-3.1.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-net-3.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\curator-client-2.7.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\curator-framework-2.7.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\curator-recipes-2.7.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\gson-2.2.4.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\guava-11.0.2.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\hadoop-annotations-2.7.0.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\hadoop-auth-2.7.0.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\hamcrest-core-1.3.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\htrace-core-3.1.0-incubating.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\httpclient-4.2.5.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\httpcore-4.2.5.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\jackson-core-asl-1.9.13.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\jackson-jaxrs-1.9.13.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\jackson-mapper-asl-1.9.13.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\jackson-xc-1.9.13.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\java-xmlbuilder-0.4.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\jaxb-api-2.2.2.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\jaxb-impl-2.2.3-1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\jersey-core-1.9.jar;C:\hadoop-2.7.0\share\had
```


Start-yarn.cmd

```
Apache Hadoop Distribution - yarn  nodemanager
26/03/23 22:04:24 INFO nodemanager.NodeManager: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NodeManager
STARTUP_MSG: host = DESKTOP-BACH1KC/192.168.1.7
STARTUP_MSG: args = []
STARTUP_MSG: version = 2.7.0
STARTUP_MSG: classpath = C:\hadoop-2.7.0\etc\hadoop;C:\hadoop-2.7.0\etc\hadoop;C:\hadoop-2.7.0\etc\hadoop;C:\hadoop-2.
7.0\share\hadoop\common\lib\activation-1.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\apacheds-i18n-2.0.0-M15.jar;C:\ha
doo-2.7.0\share\hadoop\common\lib\apacheds-kerberos-codec-2.0.0-M15.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\api-asn
1-api-1.0.0-M20.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\api-util-1.0.0-M20.jar;C:\hadoop-2.7.0\share\hadoop\common\l
ib\asm-3.2.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\avro-1.7.4.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-be
anutils-1.7.0.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-beanutils-core-1.8.0.jar;C:\hadoop-2.7.0\share\hadoop\
common\lib\commons-cli-1.2.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-codec-1.4.jar;C:\hadoop-2.7.0\share\hadoo
p\common\lib\commons-collections-3.2.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-compress-1.4.1.jar;C:\hadoop-
2.7.0\share\hadoop\common\lib\commons-configuration-1.6.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-digester-1.8
.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-httpclient-3.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-
io-2.4.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-lang-2.6.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-
logging-1.1.3.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\commons-math3-3.1.1.jar;C:\hadoop-2.7.0\share\hadoop\common\li
b\commons-net-3.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\curator-client-2.7.1.jar;C:\hadoop-2.7.0\share\hadoop\comm
on\lib\curator-framework-2.7.1.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\curator-recipes-2.7.1.jar;C:\hadoop-2.7.0\sha
re\hadoop\common\lib\gson-2.2.4.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\guava-11.0.2.jar;C:\hadoop-2.7.0\share\hadoo
p\common\lib\hadoop-annotations-2.7.0.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\hadoop-auth-2.7.0.jar;C:\hadoop-2.7.0\
share\hadoop\common\lib\hamcrest-core-1.3.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\htrace-core-3.1.0-incubating.jar;C
:\hadoop-2.7.0\share\hadoop\common\lib\httpclient-4.2.5.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\httpcore-4.2.5.jar;C
:\hadoop-2.7.0\share\hadoop\common\lib\jackson-core-asl-1.9.13.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\jackson-jaxrs
-1.9.13.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\jackson-mapper-asl-1.9.13.jar;C:\hadoop-2.7.0\share\hadoop\common\li
b\jackson-xc-1.9.13.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\java-xmlbuilder-0.4.jar;C:\hadoop-2.7.0\share\hadoop\com
mon\lib\jaxb-api-2.2.2.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\jaxb-impl-2.2.3-1.jar;C:\hadoop-2.7.0\share\hadoo\co
mmon\lib\jersey-core-1.9.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\jersey-json-1.9.jar;C:\hadoop-2.7.0\share\hadoop\co
mmon\lib\jersey-server-1.9.jar;C:\hadoop-2.7.0\share\hadoop\common\lib\jets3t-0.9.0.jar;C:\hadoop-2.7.0\share\hadoop\com
```

3. Vérification du Statut

verification de l'etat d'exécution de tout les démons (NemaNode , DataNode, RessourceManger, NodeManager) á l'aide de :

jps

```
Administrateur : Invite de commandes
C:\Windows\system32>start-dfs
C:\Windows\system32>start-yarn
starting yarn daemons
C:\Windows\system32>jps
11616 NameNode
11744 ResourceManager
19508 NodeManager
8436 DataNode
16312 Jps
C:\Windows\system32>
```

Vérification via Web

Nous avons également la possibilité d'accéder aux interfaces web.

1. NameNode Web UI

http://localhost:50070

Hadoop

Overview

Datanodes

Datanode Volume Failures

Snapshot

Startup Progress

Utilities

Overview 'localhost:9000' (active)

Started:	Mon Mar 23 22:03:50 WET 2026
Version:	2.7.0, rd4c8d4d4d203c934e8074b31289a28724c0842cf
Compiled:	2015-04-10T18:40Z by jenkins from (detached from d4c8d4d)
Cluster ID:	CID-b1c6d006-9274-42f7-8d38-42abbc46fef4
Block Pool ID:	BP-1336481334-192.168.1.7-1774299537323

Summary

Security is off.

Safemode is off.

1 files and directories, 0 blocks = 1 total filesystem object(s).

Heap Memory used 52.07 MB of 225 MB Heap Memory. Max Heap Memory is 889 MB.

Non-Heap Memory is 39.4 MB. Max Non-Heap Memory is 1 B.

Configured Capacity:	745.03 GB
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2. RessourceManager web UI

http://localhost:8088

hadoop

Cluster

About

Nodes

Node Labels

Applications

NEW

SAVING

SUBMITTED

ACCEPTED

RUNNING

FINISHED

FAILED

KILLED

Scheduler

Tools

All Applications

Logged in as: dr.who

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Memory Used	Memory Total	Memory Reserved	VCores Used	VCores Total	VCores Reserved	Active Nodes	Decommissioned Nodes	Lost Nodes	Unhealthy Nodes	Rebooted Nodes
0	0	0	0	0	0 B	8 GB	0 B	0	8	0	1	0	0	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation	Maximum Allocation
Capacity Scheduler	[MEMORY]	<memory:1024, vCores:1>	<memory:8192, vCores:32>

Show 20 entries

ID	User	Name	Application Type	Queue	StartTime	FinishTime	State	FinalStatus	Progress	Tracking UI
No data available in table										

Showing 0 to 0 of 0 entries

First Previous Next Last